



# SECRET COMMUNICATION

Cryptography, Steganography & Digital Signatures



Encrypt

Decrypt

# Secret Communication

- Cryptography, Steganography & Digital Watermarking
- Basic Lecture with Practicals
- Instructor: Misbah Rajput

# Learning Objectives

- Understand Cryptography ,steganography & watermarking
- Learn real-world applications
- Perform basic Kali Linux practicals

# What is secret Communication

- Secret communication refers to techniques used to protect information from unauthorized access.
- It ensures that messages remain confidential and secure during transmission.
- Common techniques include Cryptography, Steganography, and Digital Watermarking.

# Cryptography

- Cryptography is the science of converting readable data (plaintext) into unreadable form (ciphertext).
- It protects information using encryption and decryption techniques.

# Cryptography

## Cryptography Process Diagram



# Cryptography

- Example: Password protection and secure messaging apps.
- When you send a message on secure platforms, it is encrypted so attackers cannot read it.

# What is Steganography?

- Steganography is the practice of hiding secret data inside a cover medium like image, audio, or video.





# Example Scenario (Coded Message Using Images)

- Two illegal loggers want to communicate without writing a direct message because they are afraid that **wildlife officers** may monitor their messages.

# One day, Person A sends two images to Person B:One

-  Image of a Tree
-  Image of an Axe (cutting tool)

Although the images look harmless, **Person B understands the hidden meaning.**

| Image Sent                                                                                                                                                                         | Hidden Meaning          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
|  Tree                                                                                             | Forest / Trees location |
|  Axe                                                                                              | Cut trees               |
|  +  together | "Cut the trees tonight" |

# Interpretation

- Person B understands the message as:  
**“Go to the forest and cut the trees.”**
- **Why They Use This Method**
- No **direct text message** like “cut trees tonight”.
- If wildlife officers check the phone, they only see **normal pictures**.
- The **real meaning is hidden**, which makes it a **form of covert communication**.

# Concept for Students

- This example demonstrates **basic steganography / coded messaging**, where:
- **Normal-looking content** (images or symbols)
- hides a **secret meaning understood only by the sender and receiver**.

# Real world cases

- **Countries/Intelligence Services Known to Use This Method**
- **Germany DE**
  - Used by the intelligence agency **Abwehr** during **World War II**.
  - One famous case involved a **German spy caught by the Allies with a message written on his shaved scalp**.
- **United Kingdom GB**
  - British intelligence, especially **Special Operations Executive (SOE)**, experimented with unusual covert communication methods including invisible ink and body-hidden messages.

# How the Method Worked

- **Process (Spy Communication Technique):**
- Courier's head is **completely shaved**.
- **Secret message written on scalp** (sometimes invisible ink).
- **Hair allowed to grow** for a few weeks so the message becomes hidden.
- Courier travels to destination.
- Head shaved again → **message revealed**.

# Classic Historical Example — Tattoo Steganography (Ancient Greece)

## Story of Histiaeus and Aristagoras



# Classic Historical Example — Tattoo Steganography (Ancient Greece)

## Scenario

- Around **500 BC**, Histiaeus wanted to secretly send a message to Aristagoras.
- He shaved a slave's head and **tattooed a secret message on the scalp**.
- After hair regrew, the slave travelled safely (message hidden).
- On arrival, the receiver shaved the head again to read the message.



# Why this is Steganography???

- ✓ Message hidden inside normal appearance
- ✓ No suspicion during travel
- ✓ Requires special method to reveal
- 👉 This perfectly demonstrates “**hiding the existence of communication**”

# Modern Similar Concepts (Inspired by Tattoo Story)

- **1. Invisible Ink on Skin**
- Agents may use UV ink that becomes visible only under UV light.
-  **2. QR / Microdot Tattoos**
- Very tiny encoded tattoos (rare but conceptually similar to microdot steganography)
-  **3. Biometric Steganography**
- Hiding data inside fingerprints or medical images

# What is Watermarking?

- Digital watermarking embeds ownership or authentication data into digital media.



Original Media

Watermark  
Embed

Watermarked  
Media

# Difference

- Cryptography: Hides the content of the message.
- Steganography: Hides the existence of the message.
- Watermarking: Protects ownership of digital content.
- Steganography: Hidden communication
- Watermarking: Ownership/authentication

# Comparison Table

| Technique     | Purpose                 | Visibility             | Example                 |
|---------------|-------------------------|------------------------|-------------------------|
| Cryptography  | Protect message content | Visible but unreadable | Encrypted email         |
| Steganography | Hide message existence  | Invisible              | Message hidden in image |
| Watermarking  | Ownership protection    | Sometimes visible      | Logo in photo           |

# Applications

- Secure communication
- Banking and financial systems
- Copyright protection
- Digital forensics
- Military communication

# Application

- Secret communication is essential for protecting sensitive data.
- Cryptography, Steganography, and Watermarking each provide different security benefits.

# Real World Scenario 1

- Covert communication by spies hiding messages in images.



# Real World Scenario 2

- Copyright protection of images using watermark by photographers.

# Past Example

- Terror groups allegedly used steganography in online images (early 2000s reports).

# Tool: Steghide Practical

## 1. Steghide (★ Best for beginners)

 Already installed in Kali and very easy

### What students can do

- Hide text inside image/audio
- Extract hidden message
- Use password protection

## ◆ Basic Commands

### ✓ Hide message

⟨/⟩ Bash

```
steghide embed -cf image.jpg -ef secret.txt
```

### ✓ Extract message

⟨/⟩ Bash

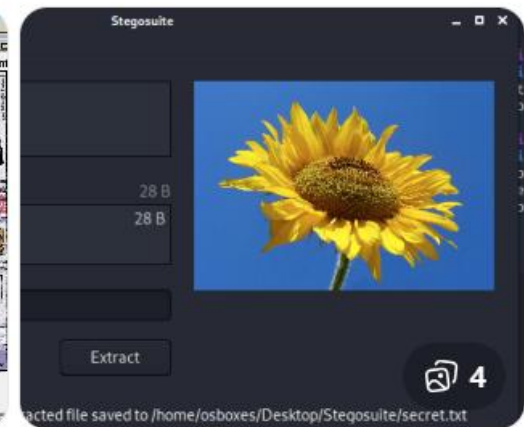
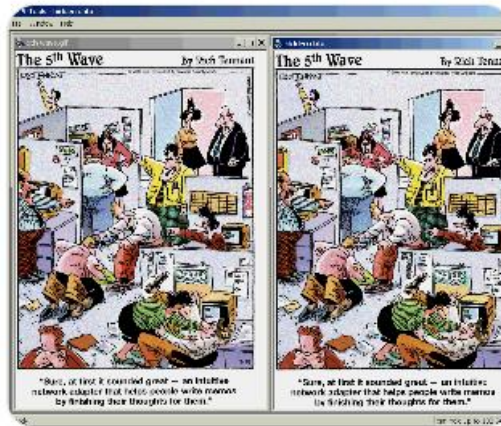
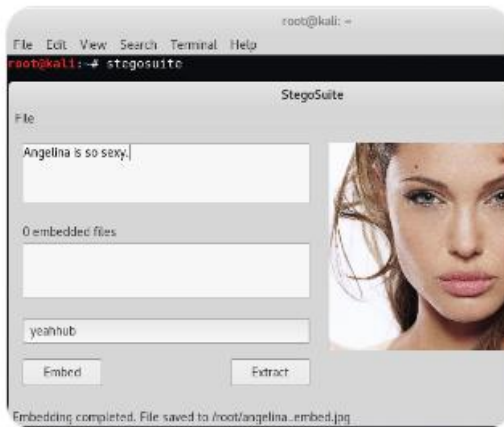
```
steghide extract -sf image.jpg
```

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# Why good for beginners

- ✓ Simple commands
- ✓ Fast results
- ✓ Password-based hiding
- ✓ Works offline

## 2. Stegosuite (★ GUI tool students love)



👉 Graphical interface (no commands needed)

# Install using kali linux

## ◆ Install (if missing)

</> Bash



```
sudo apt install stegosuite
```

## ◆ Run

</> Bash



```
stegosuite
```

# My Recommendation for Teaching Order

- Steghide → basic command concept
- Stegosuite → GUI understanding
- Outguess → advanced idea



# 3. OutGuess (Intermediate but simple)

## ◆ Hide

⟨/⟩ Bash

```
outguess -k key123 -d secret.txt image.jpg stego.jpg
```

## ◆ Extract

⟨/⟩ Bash

```
outguess -k key123 -r stego.jpg output.txt
```

# Watermark Practical

- Add watermark using ImageMagick
- `convert image.jpg -gravity south -draw "text 0,10 'ICT LAB'" watermarked.jpg`

# Class Exercise

- 1. Hide your name inside image
- 2. Exchange image with partner
- 3. Extract message

# Discussion Questions for Students

- Ethical/Unethical use?
- Difference from encryption?
- Detection challenges

# Thank You

- Questions?